Eastern Oak College of Nursing Sciences (EOCNS)

Technical Documentation

Architecture, data, publish, and extension notes for maintainers

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Live site: https://travis-global.github.io/qa_attendance/

1. Architecture overview

Client-side SPA (single page application) in HTML/CSS/JavaScript. IndexedDB database name: eocns_attendance. Canonical shared data files live under data/ in the GitHub repository and are written through a Cloudflare Worker publish endpoint (publishWorkerUrl and publishWorkerKey in data/config.json). The browser never holds the GitHub personal access token.

2. Main source files

- index.html — shell, navigation, modals, hostel and settings UI
- app.js — all business logic, reports, publish, hostel, DSA, remedial
- register.html — public student self-registration
- data/students.json, lecturers.json, sessions.json, qa_entries.json, audit.json, courses.json, timetables.json, hostel_runs.json, discipline, pending_students, programmes, levels, config.json, …
- docs/ — this handover package

3. Sync and Publish behaviour

Publish merges local and remote carefully (safePublishJson). Empty or severe shrinks of critical lists require confirmation. After publish, fingerprints mark the device clean. Smart sync pulls when the repository changes so officers are not forced to hard-refresh for routine updates. Local-authoritative rules apply for intentional deletes (e.g. students removed on purpose).

4. Hostel pipeline

Gate export → parse → computeHostelAttendance (Location ID majority for hostel, machine serial link to master list) → editable preview table (names, P/A marks, fines) → finalize download + hostel_runs master record. Train mode learns machine IDs without issuing a fee report. Unmatched users go to biometric_pending queues.

5. DSA full record

buildStudentFullRecord aggregates discipline, hostel runs, class attendance marks, QA absences, remedial, and audit. Section and full dossier print helpers are on the student full-record modal.

6. Security notes

- QA PIN in config (change via Settings flow that writes config through the worker).

- Most views blocked until isQaOfficer() session is set.
- Audit entries should carry officer name and timestamp for master edits.
- Do not commit secrets into the public repo; worker env holds GITHUB_TOKEN.

7. Extending the system

Prefer additive changes. Avoid deleting publish safety checks. After UI changes, bump app.js?v= query on index.html so browsers load the new script. Run node --check app.js before deploy. Test Publish on a copy mindset: never bulk-erase without confirmation.